

Project: Philips SmartSleep Wake-up Light

Julia Nething
nething@gatech.edu

1 INTRODUCTION

The interface that I am choosing to redesign is the [Philips SmartSleep Wake-up Light HF3520/60](#). The Philips SmartSleep is a light alarm clock that simulates the sunrise, so users wake up more gradually and in a gentler way than the typical beeping alarm clock. As the light slowly becomes brighter, the device also slowly increases the volume on some predetermined sounds or the AM/FM radio. There are many sunrise alarm clocks available, but I chose this one because it was recently chosen by Wirecutter as one of the best sunrise alarm clocks on the market (Jancer, 2021). However, there are some aspects of the design that seem odd, so it will be an interesting subject for the design life cycle. Though some light alarm clocks have a corresponding mobile app, this one does not, so the redesign will be focused to the physical device.

Unfortunately it is not possible to try out the device via the internet, but you are welcome to check out the [Amazon page](#) for the device or watch [Brandon Geekabit's product review](#) (Geekabit, 2017).



Figure 1—A picture of the Philips SmartSleep Wake-up Light HF3520/60 from its [product page on Amazon](#).

As shown in Figure 1, the device's face is primarily the light itself, though it also includes a clock and a few buttons. The majority of the buttons are on the outer rim of the light, so it is not possible to see those buttons if the user is facing

the light (two of these buttons are visible from the side in Figure 1). Common tasks that users would complete are setting the alarm for the morning, choosing a sound for the alarm, snoozing the alarm, turning off the alarm, turning off the light, turning on the "sunset" feature, changing the time (e.g. for Daylight Savings), or changing the volume. There is some personalization as well, for example users can choose from a few preset sounds or choose a station on the AM/FM radio. They can also decide the duration of the "sunrise".

2 INITIAL NEEDFINDING

2.1 Plan

For the needfinding section, I chose the top 100 reviews on Amazon after filtering for "critical reviews" and sorting by "top reviews". For each of those reviews, I read them and categorized them into a variety of different complaint types so that I can easily see which types are the most common. Figure 2 shows a portion of what this spreadsheet looks like. The first column is the number of stars (not shown), the second column is the review text, and the next 41 columns are different complaint categories. At the bottom, there is a total for the number of reviews that complained about each category. Some screenshots of all of the categories and their totals are found in *Appendix 8.1: Needfinding Raw Results*.

1	Review	Cannot replace bulb	Clock light doesn't dim, light all the time, and/or is too bright	Knocks over easily	Difficult to find the right button, oddly placed buttons (small, not intuitive)	Sunrise takes too long, wake up too early (can't change duration)	Time doesn't auto sync, update for daylight savings
	Pros: - Does a great job of waking me up in a way that allows my body to slowly adjust Cons: - Once the bulb or the alarm messes up, there's no way to fix it.						
99	I really liked this alarm clock. Even though this alarm clock is really expensive, I would have given this rating a 5/5. I saw the benefits of being able to turn the light off at night. With all that being said, I had to the rating down 2 whole stars. After a year and a half of having this light, the alarm clock suddenly no longer even turned on. The fact that this alarm clock was so expensive made me feel really cheated. When it did work, I loved it. I thought it was well worth the money. However, the fact that when it worked, it worked great would have gotten it a 5/5, but being almost a \$100 paper weight after a short time of use would have gone down to a 1/5. UPDATE: Pretty much broke after 3 months. Was great when it worked, definitely disappointed. Expected it to last longer than a 5 dollar junk alarm clock.						
100	If you're the type of person who wakes up to 7 alarms and still manages to fall back asleep, this is for you. The wait time for shipping was a little absurd. This clock is great in many ways. It has helped me wake up in a better mood. The controls are a little awkward but you get used to them. The only reason I'm giving it 4 stars is because of the shipping wait time.						
101					1		
102							
103							
	TOTAL:	2	5	10	43	2	

Figure 2—A portion of the spreadsheet used to track the top 100 critical reviews and the corresponding complaint categories.

2.2 Results

Many of the complaints focused around topics that cannot be redesigned during this design life cycle because they are outside of the scope of the interface design. These include topics such as complaints that the device quickly broke (40 reviews), that the device was cheaply made or overpriced (27 reviews), or that customer service for the device was unhelpful (4 reviews). Another com-

mon complaint was that there is no backup battery (14 reviews) and it doesn't remember any settings (so the alarm gets reset) if the device loses power (4 reviews). Those complaints will not be further delved into, although they are interesting to note.

The most common design-related complaints centered around features that were not implemented in a user-friendly way and should be improved. Users reported that it was difficult to set up and/or had a large learning curve to set up (24 reviews) and that the buttons were difficult to use for a variety of reasons: small, not intuitive, in a hard-to-reach spot, hard to know which button to press, etc. (43 reviews). The gulf of execution is high for this device, as many users complained that they had trouble using the object and some even mentioned that they gave up trying to use it. Many users also noted that the designers did not take the context into account when designing the interface. While the buttons may not be hard to press during the day, users reported that they were difficult to press when they had just woken up, a light was shining in their face, and their partner was telling them to turn it off.

Some users mentioned specific buttons that were difficult to press. They said it was hard to hit snooze (3 reviews), hard to turn off the alarm (10 reviews), and it was easy to accidentally hit the wrong button when you were trying to do something (6 reviews). Others mentioned that turning the alarm off did not turn the light off, and they felt this should be one button not two (5 reviews). They also mentioned that the device is easy to knock over (10 reviews), and when it is knocked over buttons are accidentally pressed in the process.

Users also described some features that are missing on the device that they thought were important. For example, several users were annoyed that they have to manually set the time and update it for daylight savings, because they felt that it should automatically sync (9 reviews). They also mentioned that there is no reset button on the alarm so they have to manually set the alarm every day (11 reviews), and though there is the ability to set two different alarms, the device does not keep track of the day of the week so it is not possible to have a "weekday alarm" and a "weekend alarm" (7 reviews). Users also wanted the ability to dim the brightness of the clock or even make it completely dark (5 reviews) and noted that the contrast of the clock on the face of the device made it so that it was difficult to read (3 reviews).

Several users complained about the sound options for the device. They mentioned that the default sounds were disappointing for a variety of reasons (annoying, repetitive, etc.) (11 reviews), and the only other option was AM/FM radio, no ability to play one's own music through USB or Bluetooth (8 reviews). Some also noted bad radio reception (2 reviews) and low speaker quality (3 reviews). All other complaints had 4 or fewer comments and were mostly one-offs, such as complaining that colors are not adjustable (1 review). While these complaints are also not within the scope of the interface design, they were important to note due to their prevalence in the reviews.

Overall, it seems that users need to be able to "set and forget" their alarm clocks and be able to snooze and turn off the alarm easily. They do not want to have to manually reset the alarm daily (they want a reset button) and they want to customize things like different alarms for their different weekly schedules and the brightness of the clock.

3 HEURISTIC EVALUATION

3.1 What Works Well

Much of what works well relates to the use of the buttons and the idea that many of them do what they are expected to do. The physical buttons on the outer rim of the device are also examples of *affordances*; buttons are meant to be pressed, so users know to press the buttons.

Pressing the Alarm 1 button displays the alarm time where the clock time usually is (e.g. "6:30 AM" instead of the current time). The time also blinks, which draws the user's attention to the changed time. These both help narrow the *gulf of evaluation* because users' eyes are drawn to the right place so that they can check that they have properly set the alarm time.

Pressing the "+" or "-" button next to the icon of the light not only changes the current brightness of the light emanating from the front of the device, but it also shows a number on the face (where the current time usually is) that corresponds to the brightness setting. For example, if the current brightness setting is a 10, pressing the "-" will show it going down to a 9. This narrows the *gulf of evaluation* because not only can the user tell that light's brightness is changing, but they are reassured by the number changing. This also helps users later bridge the *gulf of execution*, because in order to set the brightness level to a preferred setting

they don't have to gauge the relative brightness level every time but can instead remember a particular number that is their preferred brightness setting.

The device leverages icons instead of words for many of the buttons, for example the sun icon. This is an example of the *flexibility* principle because it makes the device usable regardless of the language that users speak or read. The sun icon also leverages *mapping* to help the user understand that pressing it will turn the light on or off; the sun icon maps well to the idea of the light. Similarly, it helps the user understand that pressing the "+" and "-" buttons next to the sun icon will change the brightness of the light.

The radio section also uses *mapping* to help the user understand what the buttons do. There is an "FM" button that turns on or off the radio itself, and on either side of that button are two megaphone sound icons that indicate that pressing one will turn down the volume and pressing the other will turn up the volume. This also uses the principle of *consistency* because these megaphone sound icons are very common on other devices to indicate volume.

There are two alarm buttons, both of them are shaped like a bell icon, which uses *mapping* to help the user understand that it is for the alarm. One of those bells has a "1" on it and the other has a "2" on it to correspond to the two alarm presets. If the user presses the Alarm 1 button, it will then show the Alarm 1 icon on the face of the device next to the clock. This helps users bridge the *gulf of evaluation* that they have successfully set the alarm, and it leverages the principle of *perceptibility* because they can immediately tell which alarms are set at any time. Similarly, if they click the button again, the icon disappears, which again helps users bridge the *gulf of evaluation* that they successfully disabled the alarm.

The device also uses the principle of *discoverability* in its main menu. Users can click the "Menu" and "Select" buttons to see what all of the options are for customizing their device. Though it is not possible to immediately undo a change, they can always press through the Menu again to change back anything they try, which means that the stakes are relatively low if they click around on the device. Users feel safe trying different buttons out because they know they can fix their mistake.

3.2 What Doesn't Work Well

Changing the alarm time is not very intuitive in my opinion. Users have to first press the alarm button, which is on the outer rim of the device (so it cannot be seen from the front). Then, they have to turn to the face of the device and look at the time that the alarm is set to, and press the "+" or "-" buttons on the front of the device (not next to the alarm button) to change the hour. To change the minute, users have to press the "Select" button on the front of the device, then press the "+" or "-" buttons. This is a *high learning curve* for a task that is pretty common for a user to complete on an alarm clock.

One button that does not use *mapping* or *consistency* appropriately is a small button on the side that looks like a snoozing emoji, as seen in Figure 3. I applied the mapping and consistency principles to deduce that this was the snooze button for the alarm clock. However, it's not; instead, pressing this button begins the "sunset" feature of the device, which means that the light will slowly dim for 20 minutes. I would have expected a sunset icon instead for this feature so that it would not be confused with snoozing the alarm.



Figure 3—A picture of the button that looks like a snoozing emoji on the Philips SmartSleep Wake-up Light HF3520/60 from its [product page on Amazon](#).

Changing the duration of the sunrise is very difficult and would be almost impossible to find without using the manual. The user has to hold down the Alarm 1 and the Alarm 2 buttons at the same time until an integer shows where the clock time usually is displayed. That is the duration of the sunrise in minutes, and the user can then press the "+" or "-" button to change that time. There is

no indication that holding down both buttons would be necessary. In addition, there is no reason why the sunset duration couldn't be part of the regular alarm menu, which also allows for users to set the brightness of the light or the volume of the alarm sound. This goes against the principle of *discoverability* because there is no reason for users to hold down random combinations of buttons in order to discover hidden menu options. This also shows *expert blindspot*; an expert designer may think that that makes sense to hold down those buttons, but a novice would not think to do that to change the duration, and they may therefore conclude that it is not possible to change the duration because they have already clicked through all of the menu options and not found it.

The *gulf of execution* is higher because of the buttons on the outer rim, which are not visible to the user if they are facing the device. This means that users have to keep in working memory which buttons are available and where they are positioned around the device in order to reach to press them. Otherwise, they are forced to turn the entire device down or on its side to make even a small adjustment. This in turn widens the *gulf of evaluation* because they have to turn the device back to see if they are changing what they want to change. For example, if the user wants to increase the volume, they have to turn the device down to press the correct button, then they have to turn the device back up to see what the current numeric volume level is. This also means that if your cognitive load is higher (for example, if you just woke up), and you want to turn off the alarm as quickly as possible, you have to either remember where the button is located on the outer rim or pick up and turn around the device while looking for the correct button.

The device also breaks the principles of *ease, comfort, and flexibility* in terms of how users with disabilities can use the device. For example, the buttons on the outer rim of the device are impossible to see when facing the device. Users have to reach above or to the side of the device to press those small buttons, which may be difficult for people with hand tremors or mobility issues. It is not easy nor is it comfortable to perform this reach-around action, and it is not flexible for users with varying abilities. In addition, the clock display has low contrast with its background, making it difficult for people with visibility issues to read the display, which also does not take into account users with varying abilities.

Finally, the device violates the principle of *tolerance* due to its snooze button and its alarm off button. The only way to turn the alarm off is to press the alarm

button, but that turns the alarm off permanently (does not reset it for the next morning). If the user did not know that or forgets to turn it back on every day, there is a chance that user will not wake up in morning and be late to an important event. In addition, the snooze button is not marked and is accomplished by tapping the front of the device. Because it is so hard to find the correct button on the outer rim, it is very likely that users will accidentally tap the snooze button while they are trying to pick up the device to find the off button. The device is intolerant of this error and will snooze the alarm, which may make the user think that they successfully turned it off when they did not. This may then aggravate the user or their partner, who wants the alarm off.

4 INTERFACE REDESIGN

For the purpose of this redesign, I am only focusing on the main tasks that a user needs to complete: set up an alarm, turn off an alarm, snooze an alarm, and change the brightness or volume settings. I will not focus on other settings (such as brightness of the display) or changing the sounds on the device, because that simply expands the scope too much. The general design for those features would be very similar to how an alarm is edited in Figures 10 and 11. All blue text in the figures are for reference, not part of the design.

4.1 Overall Device Design

This is shown in Figure 4.

4.2 Changing the Brightness or Volume

This is shown in Figures 5 and 6.

4.3 Snoozing or Turning Off an Alarm

This is shown in Figures 7 and 8.

4.4 Navigating to Home/Menu

This is shown in Figure 9. Though all three options are shown (alarm, sound, and system settings), only alarm settings are delved into further below. The sound and system settings are outside the scope of this redesign.

4.5 Setting an Alarm

This is shown in Figures 10 and 11.

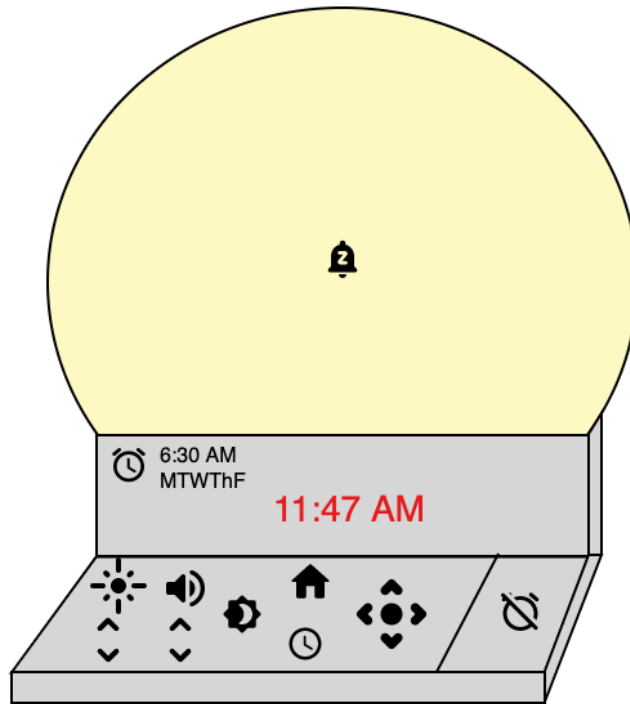


Figure 4—A mockup of the redesign of the device as a whole. The light is a semicircle above the clock section, and then a separate menu area is perpendicular to the clock/light part. The clock section is just a display (no buttons), while the menu area is all buttons: light up, light down, sound up, sound down, sunset, home/menu view, clock view, navigation buttons (up/down/left/right/select), and alarm off. Users can also tap the light itself (where the snooze icon is) to snooze the alarm.

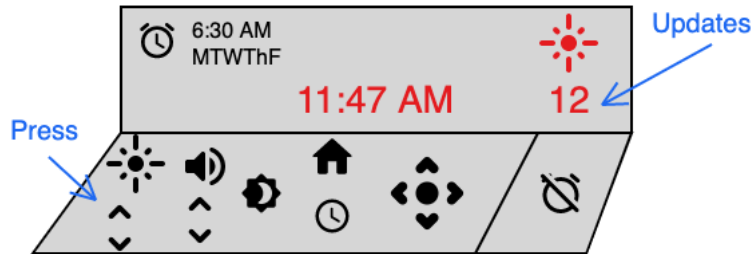


Figure 5—When the up or down arrow next to the sun icon is pressed, the corresponding numeric value changes in the display section next to the clock, along with the sun icon. This display data shows while the user is pressing the buttons and then disappears a couple seconds after the last button press.

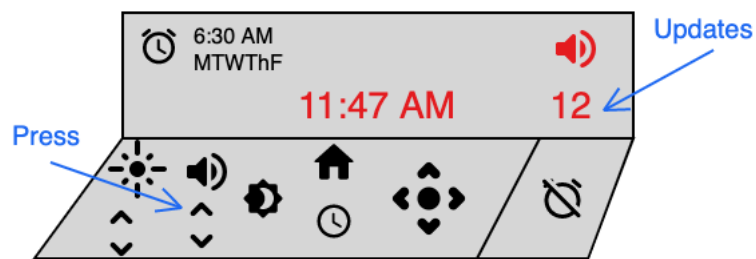


Figure 6—When the up or down arrow next to the sound icon is pressed, the corresponding numeric value changes in the display section next to the clock, along with the sound icon. This display data shows while the user is pressing the buttons and then disappears a couple seconds after the last button press.

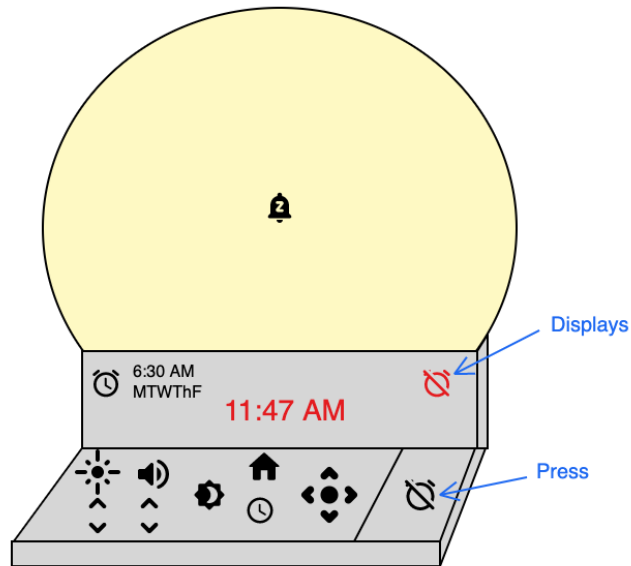


Figure 7—When the button that has the crossed-out alarm icon on it is pressed, the same icon shows in red in the display section next to the clock. This indicates that the alarm has been successfully reset. In this design, the alarm is just reset, so it will still go off on its next scheduled day unless edited in the settings.

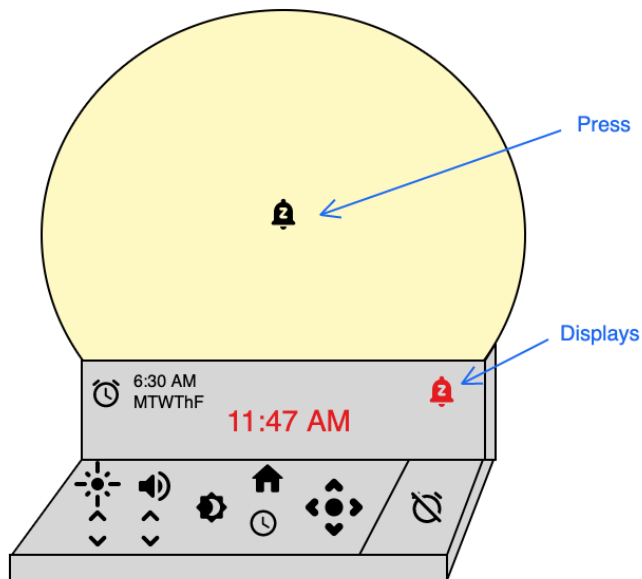


Figure 8—When the face of the light itself is pressed (where the snooze icon is), the snooze icon shows in red in the display section next to the clock. This indicates that the alarm has been successfully snoozed.

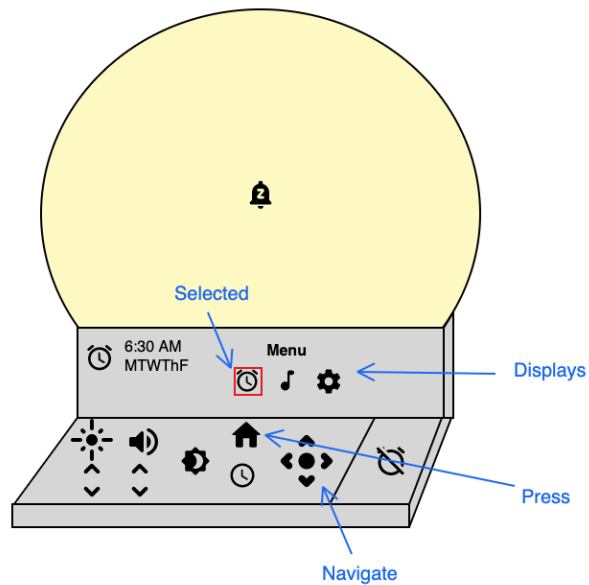


Figure 9—Pressing the "home/menu" icon displays the menu where the clock usually is (pressing the "clock" icon would change it back). This shows three options: alarm, sound, or system settings (designated by the different icons). The red box indicates which item is currently highlighted. The arrow buttons and center select button are used to navigate the menu.

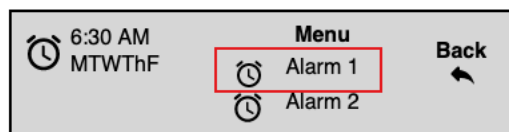


Figure 10—After navigating to the menu and clicking on the alarm icon, the display shows a list of the alarms to choose from. The one surrounded by a red box is currently highlighted, and then users can select that with the center select button to edit that particular alarm. The "back" button would take the user back to the main menu.

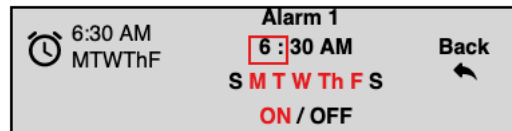


Figure 11—After choosing the particular alarm to edit, users are taken to an edit menu for that alarm, which includes the time, recurring day of the week, and whether it is currently turned on or off. Each section is editable (the hour, the minute, each individual day of the week, etc.) and users can use the arrow buttons to move between them to edit each one. The red color indicates that it is selected (i.e. a red "M" means that Monday is selected as a recurring day of the week). If an alarm is turned from on to off, it will be removed from the permanent display on the left, and if it is turned on it will be added to the permanent display on the left.

5 INTERFACE JUSTIFICATION

5.1 Overall Device Design

For the overall device design, I kept many attributes that make the design good, while mitigating some of the attributes that make it difficult to use. For example, the buttons around the outer rim in the original design are *affordances* that help the user to understand to press the buttons in order to use them. That part of the design was kept, so almost all of the interactions on the device are done by pressing buttons. However, the new design leverages the principles of *ease*, *comfort*, and *flexibility* by adding more weight to the base of the device so it is not easy to knock over accidentally. This helps users with mobility issues or hand tremors not have to worry about accidentally knocking the device over, which gives flexibility to users of different abilities. This is described in more detail in the next section, but I also moved the buttons from the outer rim, which were impossible to see when facing the device, to the base of the device where they are easy to see.

5.2 Changing the Brightness or Volume

One aspect of the original design that was easy to use was the up and down arrows next to the light icon for adjusting brightness and the corresponding arrows next to the megaphone icon to adjust the volume. These both use *mapping* to help the user understand that those buttons control the light and sound, re-

spectively. Changing the light or sound also displayed a corresponding numeric value which helped narrow the *gulf of evaluation*, so users knew that they were affecting those settings. It also helped users with the *gulf of execution* because they could remember the numeric value that they wanted to set it to instead of having to manually compare volumes or brightness levels. These are all positive attributes that I kept in the redesign.

For the redesign, I wanted to make improvements by leveraging the principles of *ease and comfort* to make the buttons easier to reach, especially for users with mobility issues. This in turn lessens the user's cognitive load by not requiring them to keep the location of the buttons in working memory, because the buttons in the original design were not visible when facing the device. By moving the buttons to the base of the device in the redesign, users can easily press those buttons and not have to remember where they are.

5.3 Snoozing or Turning Off an Alarm

One aspect of the original design that was good was that users could snooze the alarm by tapping the face of the light itself. This leveraged the principle of *ease* because the face of the light is a large and easy target for users to aim at. Users noted in the needfinding stage that when they have just woken up in the morning it is difficult for them to find the right button to turn off the alarm, and it was easy for them to accidentally press other buttons while they were trying to hit the snooze button. In this redesign, I have kept the large target to hit the snooze button, which helps narrow the *gulf of execution*, and I moved all other buttons to the perpendicular base of the light, which means that it is not as easy to accidentally tap another button while attempting to tap the snooze button. I wanted to add a similar large button to turn off the alarm that would be easy for users to press, even when they have a high cognitive load (they are just waking up, a light is shining in their face, and their partner is telling them to turn the alarm off). That's why the new "alarm off" button is much bigger than all of the other buttons, and it's on the edge of the base so users can feel that it is the correct button even when they can't see it properly because a light is shining in their face. This also narrows the *gulf of execution* because they can find the button without needing to look at it, and it's large so they don't need to be as precise when pressing it. This also leverages the *ease, comfort, and flexibility* principles because it makes it easier for users with mobility issues or hand tremors to turn off the alarm quickly (precision is not necessary).

In addition, the redesign narrows the *gulf of evaluation* by showing icons on the display section that correspond to the button that was just pressed. After tapping the snooze button, the red snooze icon will appear in the display, and after tapping the alarm off button the corresponding icon will also appear in the display. This helps users understand that they have accomplished their goal.

5.4 Navigating to Home/Menu

The original design had four buttons on its face: "-", "menu", "select", and "+". While these buttons are common buttons to have on a device, usually navigating through menus is done through a typical arrow formation with a select button in the center of the arrows. This button configuration is common on regular alarm clocks, TV remotes, etc. Therefore, I leveraged the *consistency* principle by using that same configuration in the redesign for users to navigate through the menu. In addition, by using the principle of *mapping*, I used the home icon to represent the home/menu view and the clock icon to represent the current time view.

5.5 Setting an Alarm

Setting an alarm is a task that users typically do not need to do frequently. Most often, users will set 1-2 recurring alarms and then not have to change the settings again unless they have a major schedule change. Therefore, I leveraged the principle of *simplicity* and removed the individual alarm buttons from the device itself and just added those options to the settings menu. This also makes it easier for users to find all of the options available to them for editing each alarm, which leverages the *discoverability* principle.

In the original design, the display view showed an icon for which alarm(s) were set, which I wanted to keep in this design because it helps narrow the *gulf of evaluation* and leverages the principle of *perceptibility* because users can immediately see which alarms are currently set. However, I felt that simply showing "Alarm 1" or "Alarm 2" is not clear enough. If a user does not remember which one is Alarm 1 versus which one is Alarm 2, they will have to click through various buttons to see what the settings are for each alarm. Therefore, in this redesign, the details of the currently set alarm are shown directly on the display view to the left of the current time. It shows not only the time that the alarm is set for, but also which days of the week that it will repeat on. The previous design did not include the day of the week, so this is a new feature for the redesign.

If the user edits an alarm that is already set, it will automatically update the changed details on the left side of the display. This narrows the *gulf of evaluation* because users can immediately see the effect that their actions have made on the currently set alarm. If they turn off that alarm, it will remove it from the left side of the display as well.

6 EVALUATION PLAN

6.1 Overview

The evaluation plan that I would use to evaluate this interface would be an empirical evaluation. The reason is that with the exception of the addition of the day-of-the-week recurring alarm schedule, all the changes made in this redesign are just improving existing functionality and not adding any new features. While it would be ideal if the prototype could be made into a 3D prototype for the purpose of the evaluation, the process for each task can still be evaluated by comparing the wireframe prototype with the original device.

6.2 Independent and Dependent Variables

For this evaluation, I will have participants test out a variety of tasks on the original device and on the redesigned prototype. The control condition, or independent variable, is the device itself (the original device versus the prototype). The experimental conditions, or dependent variables, will be how many steps or button presses it takes for users to accomplish a variety of tasks, which are described in the Experimental Method section below.

6.3 Control and Experimental Group

For this evaluation, I will use a Within Subjects design in order to help limit the number of participants needed for the evaluation. For each participant, it will be randomly decided which treatment (the original device or the prototype) they will receive first, in order to limit lurking variables.

6.4 Null and Alternative Hypotheses

The null hypothesis is that there is no difference in the number of button presses the user has to make to accomplish each task on the original device and on the prototype. The alternative hypothesis is that there is statistically significant difference in the number of buttons the user has to press in order to accomplish

each task on the original device and the prototype.

6.5 Experimental Method

For this analysis, participants will complete certain tasks, and the data generated will be the number of steps or button clicks that were necessary to complete each task. Sometimes "number of steps" is more meaningful than "number of button clicks" because a step may include turning the device to see the button first.

Here are the tasks that participants will complete:

- Snooze an alarm
- Turn off an alarm
- Create a new alarm set for 6 AM tomorrow
- Change the time (e.g. the hour) on an alarm
- Evaluate if the correct alarm is turned on for tomorrow
- Change from one's weekday alarm to one's weekend alarm
- Change the brightness of the light
- Change the volume of the sound

6.6 Planned Statistical Analysis

The number of clicks is ratio data, so the analysis I will perform on this data is a Student's t-test. This will help me evaluate if the null or alternative hypothesis is correct.

6.7 Lurking Variables

One lurking variable is which treatment the participant receives first. Whichever treatment they receive second might be automatically easier for them because they are more familiar with the task and the context of the problem.

Another lurking variable is the familiarity the user has with alarm clocks, and in particular light alarm clocks, in general. This is especially true if they have not used a physical alarm clock in a while (e.g. if they use a smartphone app), they might be confused how to program it at all, while a user who does have a physical alarm clock may be more likely to know what to do.

7 REFERENCES

- [1] Geekabit, Brandon (2017). *Philips HF3520 Wake-Up Light Review Setup*. URL: <https://www.youtube.com/watch?v=Z9LqQEn96Ew>.
- [2] Jancer, Matt (Mar. 2021). *The 6 Best Sunrise alarms to help you rise and shine*. URL: <https://www.wired.com/gallery/best-sunrise-alarm-clocks/>.

8 APPENDICES

8.1 Needfinding Raw Results

These are screenshots of all of the categories and their totals from going through the top 100 critical reviews on Amazon.

Stress	Reactive	Control regions below	Click light doesn't stop light on the line, or/for it is too bright	Knows over easily	Difficult to be sure about bottom line and incline	Remains late too long, even if he early (can't change direction)	Time doesn't suit me too long for daylight brightness	No backup light	Easy to misinterpretly hit someone or other bodies	Hard to hit someone	Hard to turn alarm off	Default sounds are misleading and not something operative	Breaks, usually on weekend	Doesn't work days have to change on weekend	Difficult to see the alarm on large leaving cars
Buy the alarm clock if you want your spouse/partner/girlfriend to wake you because they cannot operate it and you can't turn the alarm off.															
Buy the alarm clock if you want your spouse/partner/girlfriend to wake you because they cannot operate it and you can't turn the alarm off.															
1 On the bright side there is a pay off for months of use I'll save the morning light quality and the sound being right															
1 After about 6 months, I decided to stop working. The warranty information listed with the product brings up a general list of things, none of which I thought I was just going for until a month. Then I went to change some settings (alarm sound) and the sequence for input a longer delay to 3 by speed sensitivity. I used, used reference 2 was, when I wrote the result is 1 item.															
1 I loved the alarm clock for almost a year, then it abruptly stopped working. The battery buttons were completely unresponsive. I changed it to be															
1 I read the reviews that said it's not operable in any way but I bought it in spite of that. But now that more actually did, I can recommend this in good															
2 It's off a week or so now.															
1 After six months, when I got to my scheduled wake-up time, the light suddenly went off. Then, I unplugged the machine and plugged it back in. It															
1 I contacted Philips via online chat (as I was a few years old), I was told they needed to replace it and I was sent a small receipt to a specific email as I															
1 I actually designed several alarm clocks and chose this one. I used for 5 days I was very good at day one. The problem is that I don't really see															
1 I was just about to go to bed when I saw and have found the only awful side effect is that I realized my bed was still empty, clearing so much on my r															
1 I was just about to go to bed when I saw and have found the only awful side effect is that I realized my bed was still empty, clearing so much on my r															
1 This is probably the worst design for an alarm clock I've ever encountered. For the price you'd expect much more. The controls are not intuitive at a															
1 This is probably the worst design for an alarm clock I've ever encountered. For the price you'd expect much more. The controls are not intuitive at a															
1 This is probably the worst design for an alarm clock I've ever encountered. For the price you'd expect much more. The controls are not intuitive at a															
1 This is probably the worst design for an alarm clock I've ever encountered. For the price you'd expect much more. The controls are not intuitive at a															
1 This is probably the worst design for an alarm clock I've ever encountered. For the price you'd expect much more. The controls are not intuitive at a															
1 This is probably the worst design for an alarm clock I've ever encountered. For the price you'd expect much more. The controls are not intuitive at a															
1 This is probably the worst design for an alarm clock I've ever encountered. For the price you'd expect much more. The controls are not intuitive at a															
1 This is probably the worst design for an alarm clock I've ever encountered. For the price you'd expect much more. The controls are not intuitive at a															
1 This is probably the worst design for an alarm clock I've ever encountered. For the price you'd expect much more. The controls are not intuitive at a															
1 This is probably the worst design for an alarm clock I've ever encountered. For the price you'd expect much more. The controls are not intuitive at a															
1 This is probably the worst design for an alarm clock I've ever encountered. For the price you'd expect much more. The controls are not intuitive at a															
1 This is probably the worst design for an alarm clock I've ever encountered. For the price you'd expect much more. The controls are not intuitive at a															
1 This is probably the worst design for an alarm clock I've ever encountered. For the price you'd expect much more. The controls are not intuitive at a															
1 This is probably the worst design for an alarm clock I've ever encountered. For the price you'd expect much more. The controls are not intuitive at a															
1 This is probably the worst design for an alarm clock I've ever encountered. For the price you'd expect much more. The controls are not intuitive at a															
1 This is probably the worst design for an alarm clock I've ever encountered. For the price you'd expect much more. The controls are not intuitive at a															
1 This is probably the worst design for an alarm clock I've ever encountered. For the price you'd expect much more. The controls are not intuitive at a															
1 This is probably the worst design for an alarm clock I've ever encountered. For the price you'd expect much more. The controls are not intuitive at a															
1 This is probably the worst design for an alarm clock I've ever encountered. For the price you'd expect much more. The controls are not intuitive at a															
1 This is probably the worst design for an alarm clock I've ever encountered															

Figure 12—The first portion (of three) of the spreadsheet used to track the top 100 critical reviews and the corresponding complaint categories.

[illegible]

Figure 13—The second portion (of three) of the spreadsheet used to track the top 100 critical reviews and the corresponding complaint categories.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200	201	202	203	204	205	206	207	208	209	210	211	212	213	214	215	216	217	218	219	220	221	222	223	224	225	226	227	228	229	230	231	232	233	234	235	236	237	238	239	240	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255	256	257	258	259	260	261	262	263	264	265	266	267	268	269	270	271	272	273	274	275	276	277	278	279	280	281	282	283	284	285	286	287	288	289	290	291	292	293	294	295	296	297	298	299	300	301	302	303	304	305	306	307	308	309	310	311	312	313	314	315	316	317	318	319	320	321	322	323	324	325	326	327	328	329	330	331	332	333	334	335	336	337	338	339	340	341	342	343	344	345	346	347	348	349	350	351	352	353	354	355	356	357	358	359	360	361	362	363	364	365	366	367	368	369	370	371	372	373	374	375	376	377	378	379	380	381	382	383	384	385	386	387	388	389	390	391	392	393	394	395	396	397	398	399	400	401	402	403	404	405	406	407	408	409	410	411	412	413	414	415	416	417	418	419	420	421	422	423	424	425	426	427	428	429	430	431	432	433	434	435	436	437	438	439	440	441	442	443	444	445	446	447	448	449	450	451	452	453	454	455	456	457	458	459	460	461	462	463	464	465	466	467	468	469	470	471	472	473	474	475	476	477	478	479	480	481	482	483	484	485	486	487	488	489	490	491	492	493	494	495	496	497	498	499	500	501	502	503	504	505	506	507	508	509	510	511	512	513	514	515	516	517	518	519	520	521	522	523	52
--	---	---	---	---	---	---	---	---	---	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	----